



Pengembangan Pipeline Konstruksi Basis Pengetahuan pada Sistem OMNI-RAG Multi-Domain Berbasis Plugin pada Studi Kasus Peraturan Perundang-undangan dan User Manual

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01 Why is multi-domain RAG difficult?

Different document domains require different knowledge construction strategies.

INDONESIAN LEGAL DOCUMENTS

PRODUCT USER MANUALS

Hierarchical regulation -> article -> paragraph versus procedure -> table -> product term.

02 What is the objective?

Develop an extensible knowledge construction pipeline for multi-domain RAG without modifying the core orchestrator when a new strategy or domain is introduced.

new strategy = new plugin

contracts keep the orchestrator stable

03 What was developed?

A shared canonical representation turns one parsed document into interchangeable retrieval projections.

4
chunkers

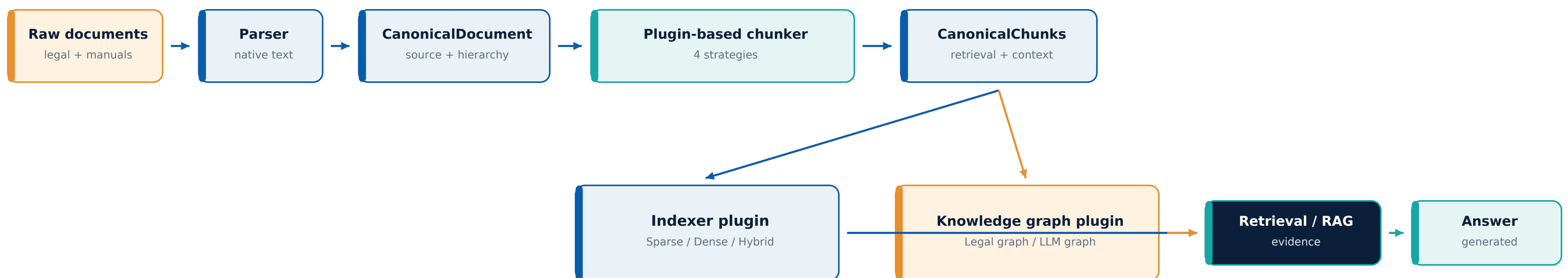
3
indexers

2
graph strategies

Canonical artifact + provenance + plugin contracts

04 Proposed architecture - one canonical path, many strategies

CONTROL LAYER | Profile | Manifest | Provenance | Compatibility contracts



Same chunks -> different projections. New strategy = new plugin, not a core-orchestrator rewrite.

05 Experimental setup - two domains, one controlled protocol

DATASETS

9

Netgear manuals

1,400 pages | 100 bilingual slots

256

legal documents

49 reference | 207 noise | 25 questions

PROTOCOL

K = 30

sweep K = 5-60

generation K = 10

Recall primary

Supporting metrics: Precision, MRR, NDCG
Generation: correctness + faithfulness

CONFIGURATION SPACE

4 chunkers

3 indexers

2 graph strategies

OFAT: change one component, hold the rest fixed
Direct retrieval path: reranker and graph disabled

06 Experimental evidence - structure and semantics change coverage

Preserving structure improves retrieval coverage



+0.055 Recall@30

Semantic dense retrieval leads recall



Dense vs sparse: +0.365 recall

Domain-aware structure transfers to legal text



Best legal Recall@30: 0.6854

Downstream generation remains strong

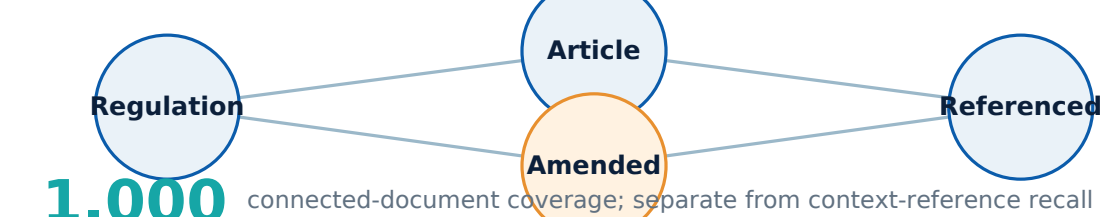
0.788 / 0.958

manual correctness / faithfulness

0.690 / 0.972

legal correctness / faithfulness

Graph expands connected coverage



07 What did we learn?

01 Structure matters

Structure-aware chunking increases context-reference coverage across both tested domains.

02 Semantic retrieval matters

Dense indexing substantially improves recall compared with the sparse baseline in the tested configurations.

03 Modularity matters

Canonical artifacts and plugin contracts enable new strategies without modifying the core orchestrator.

OMNI-RAG separates domain-specific knowledge construction from the core pipeline - making multi-domain RAG more extensible, traceable, and experimentally configurable.